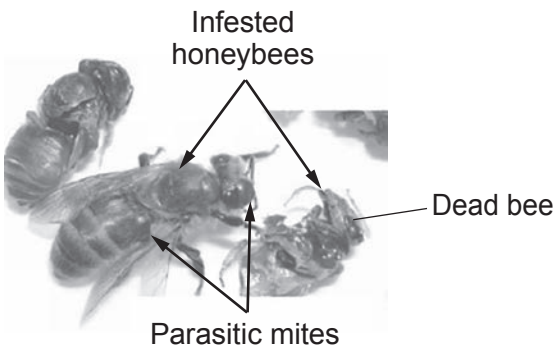


10. *Varroa destructor* is a parasitic mite of honeybees. A colony of honeybees affected by the parasite becomes weakened and eventually dies out. The parasitic mite originated in Asia, where it is a pest of the Asian honeybee, *Apis cerana*. It has spread across most continents, arriving in the UK in the early 1990s and affecting the native European honeybee, *Apis mellifera*.



- (a) Use your knowledge of classification to suggest why the parasitic mite was able to spread from the Asian to the European honeybee. [2]

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In 2014-2015 there were 1650 registered beekeepers in Wales who kept bee colonies, each of which is one beehive. The National Bee Unit recommends that honeybees should be treated with pesticides to try and reduce the numbers of the parasitic mite in beehives.

In recent years many beekeepers in North West Wales have stopped treating their bees with pesticides. These beekeepers believe that there is no difference in colony losses between bees treated with pesticides and bees not treated with pesticide.



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The table below shows Winter Losses (the number of colonies that died out) between 2010-2015 in North West Wales.

Season	Number of beekeepers in the survey		Total number of colonies	Number of treated colonies	% winter loss in treated colonies	Number of colonies not treated	% winter loss in not treated colonies
	Treating	Not treating					
2010-2011	10	5	71	44	27	27	11
2011-2012	11	31	355	180	8	175	7
2012-2013	8	46	251	75	41	176	32
2013-2014	12	55	396	81	9	315	6
2014-2015	17	65	500	97	8	403	8

(b) What percentage of beekeepers in Wales took part in this survey in 2014-2015? [2]

percentage of beekeepers = %

(c) (i) Use the information in the table to state **one** conclusion about the effect of the pesticides on winter losses of bees. [1]

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(ii) Give **two** reasons why some beekeepers across the UK would argue that the strength of evidence in this survey is weak. [2]

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(d) Early attempts at control of the parasitic mite involved using the chemical, pyrethroid. By the late 1990s the parasitic mite developed resistance to the chemical and beekeepers were advised to use other chemicals to control the pest.

Explain why, over time, the pyrethroid chemical became less effective at killing the parasitic mite. [4]

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